

Please check the examination details below before entering your candidate information

Candidate surname					Other names				
Centre Number					Candidate Number				

Pearson Edexcel International GCSE

Wednesday 13 May 2026

Afternoon (Time: 1 hour 30 minutes)

Paper reference **4MB1/01**

Mathematics B

PAPER 1



You must have: Ruler graduated in centimetres and millimetres, protractor, pair of compasses, pen, HB pencil, eraser, calculator. Tracing paper may be used.

Total Marks

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- **Calculators may be used.**

Information

- The total mark for this paper is 100.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Check your answers if you have time at the end.
- Without sufficient working, correct answers may be awarded no marks.

Turn over ►

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Answer ALL TWENTY SIX questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

- 1** Solve the inequality $6x < 20 - 2x$

.....
(Total for Question 1 is 2 marks)

- 2** Factorise fully $8ab - 28a$

.....
(Total for Question 2 is 2 marks)

- 3** Find the Lowest Common Multiple (LCM) of 60 and 84
Show all your working.

.....
(Total for Question 3 is 2 marks)



- 4 (a) Write down the number from the list that is a natural number.

-3 $\frac{1}{6}$ 4 4.5 π

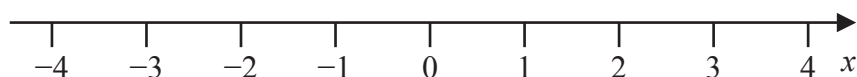
.....
(1)

- (b) Write down an irrational number between 1 and 2

.....
(1)

(Total for Question 4 is 2 marks)

- 5 Show on the number line the inequality $-2 < x \leq 3$



(Total for Question 5 is 2 marks)

- 6 Each time Ishan plays a game, the probability that he wins is 0.64
Ishan plays the game 50 times.

Calculate the expected number of games Ishan does **not** win.

.....
(Total for Question 6 is 2 marks)



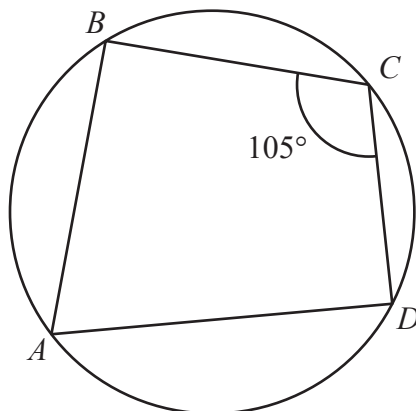


Diagram **NOT**
accurately drawn

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In the diagram A , B , C and D are points on a circle.

$$\angle BCD = 105^\circ$$

Find the size, in degrees, of $\angle BAD$

Give a reason for your answer.

(Total for Question 7 is 2 marks)



- 8 The price of a yacht is reduced by 16% in a sale.

Leda pays \$15 700 of the sale price when she orders the yacht.

She pays the remainder of the sale price in 10 equal payments of \$2000

Calculate the price of the yacht before the sale.

\$.....

(Total for Question 8 is 3 marks)

- 9 Write as a single fraction in its simplest form

$$\frac{4x - 3}{3} - \frac{2x + 5}{2}$$

Show clear algebraic working.

(Total for Question 9 is 3 marks)



10 Zahra records the time, t hours, some people spend watching television each day.

Her results are shown in the table below.

Time (t hours)	Frequency
$0 \leq t < 2$	13
$2 \leq t < 4$	20
$4 \leq t < 7$	27

She draws a histogram for the information in the table.

In the histogram

the bar for the class interval $4 \leq t < 7$ is a square with sides of length 4.5 cm

the bar for the class interval $2 \leq t < 4$ is a rectangle of width w cm and height h cm

Find the value of w and the value of h

$w =$

$h =$

(Total for Question 10 is 3 marks)



11 Rima sells green pens, red pens and blue pens only.

On Saturday, a total of 200 pens are sold by Rima.

The ratio of the number of green pens sold to the number of red pens sold is 3:5

The ratio of the number of green pens sold to the number of blue pens sold is 2:3

Calculate the number of **red** pens sold by Rima on Saturday.

(Total for Question 11 is 3 marks)



12 $x = 4 \times 10^{-105}$ and $y = 3.2 \times 10^{64}$

(a) Calculate xy

Give your answer in standard form.

.....
(2)

(b) Calculate $\frac{y}{x}$

Give your answer in standard form.

.....
(2)

(Total for Question 12 is 4 marks)



- 13 x is directly proportional to $\sqrt{w}$
 y is inversely proportional to w

$$x = 2 \text{ when } y = 16$$

Find a formula for y in terms of x

.....
(Total for Question 13 is 4 marks)



- 14 Quadrilateral A is transformed to quadrilateral B under a reflection in the y -axis.
Quadrilateral B is transformed to quadrilateral C under an enlargement with centre $(0, 0)$ and scale factor 4

Quadrilateral C is the image of quadrilateral A under the matrix N

Find the matrix N

Show all your working.



(Total for Question 14 is 4 marks)



15 Make x the subject of

$$\frac{2}{x} - \frac{2}{x+4} = \frac{8}{y} \quad \text{where } x \text{ and } y \text{ are positive integers.}$$

Give your answer in the form $a + \sqrt{y+b}$ where a and b are integers.

Show clear algebraic working.

(Total for Question 15 is 4 marks)



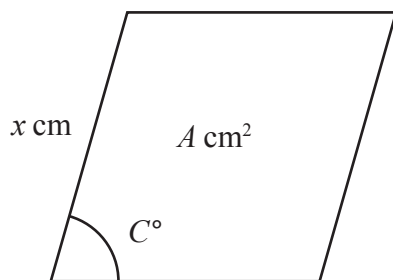


Diagram **NOT**
accurately drawn

The diagram shows a rhombus of area $A \text{ cm}^2$ in which

$x = 8.4$ to 1 decimal place

$A = 50$ to 1 significant figure

Given that the angle marked C° is an acute angle,

find the upper bound, to 3 significant figures, of C

Show all your working.

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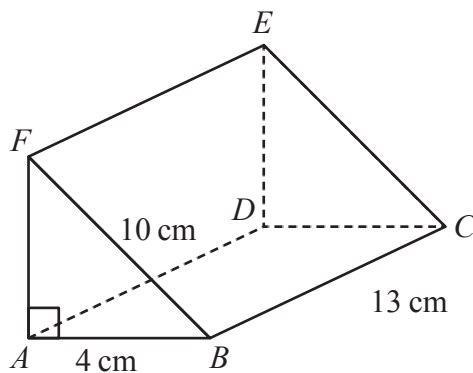
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(Total for Question 16 is 4 marks)



Diagram **NOT**
accurately drawn



The diagram shows a right triangular prism $ABCDEF$

$$AB = 4 \text{ cm} \quad BC = 13 \text{ cm} \quad FB = 10 \text{ cm} \quad \angle FAB = 90^\circ$$

Calculate the volume, in cm^3 to the nearest integer, of the triangular prism.

..... cm^3

(Total for Question 17 is 4 marks)



18 The n th term of a sequence is given by $10 + 6n$

- (a) Find the sum of the 3rd term and the 7th term of this sequence.

.....
(2)

For a different sequence the first five terms are

7 11 15 19 23

- (b) Write down a formula for the n th term of this sequence.

.....
(2)

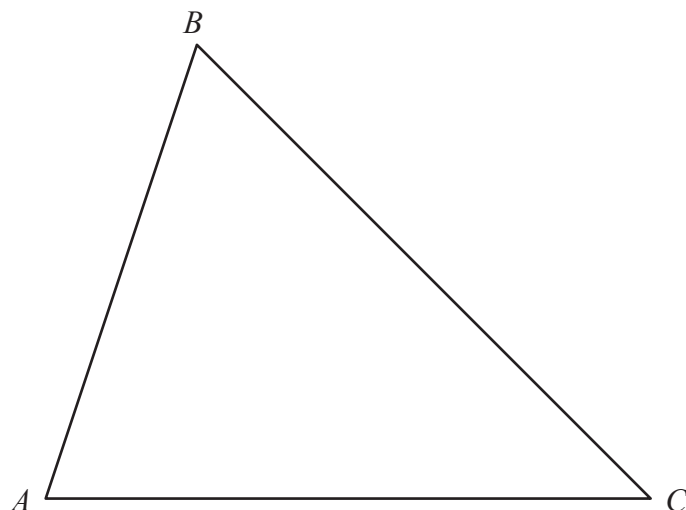
Ian says there are no numbers that are in both the sequence in part (a) and the sequence in part (b).

- (c) Explain why Ian is correct.

(1)

(Total for Question 18 is 5 marks)





The diagram shows triangle ABC

- (a) Construct the locus of all points inside the triangle that are 7 cm from C (1)
 - (b) Construct the locus of all points inside the triangle that are 3 cm from AC (1)
 - (c) Using ruler and compasses only and **showing all your construction lines**, construct the locus of all points inside the triangle that are equidistant from AB and AC (2)
- The region R consists of all the points inside the triangle that are less than 7 cm from C , nearer to AB than to AC and more than 3 cm from AC
- (d) Show, by shading, the region R (1)

(Total for Question 19 is 5 marks)

20 (a) Without using a calculator and showing your working clearly, express

$$\sqrt{700} - \sqrt{343}$$

in the form $\sqrt{a}$ where a is an integer.

.....
(2)

(b) Without using a calculator and showing your working clearly, express

$$\frac{\sqrt{3} - \sqrt{5}}{\sqrt{3} + \sqrt{5}}$$

in the form $c + \sqrt{d}$ where c and d are integers.

.....
(3)

(Total for Question 20 is 5 marks)



21 **A** and **B** are two mathematically similar solids.

The volume of **A** is $324\,000\text{ cm}^3$

The volume of **B** is $12\,000\text{ cm}^3$

The total surface area of **A** is 3^{x+6} cm^2

The total surface area of **B** is 9^x cm^2

Given that the height of **B** is 5^x cm

find, in cm, the height of **A**

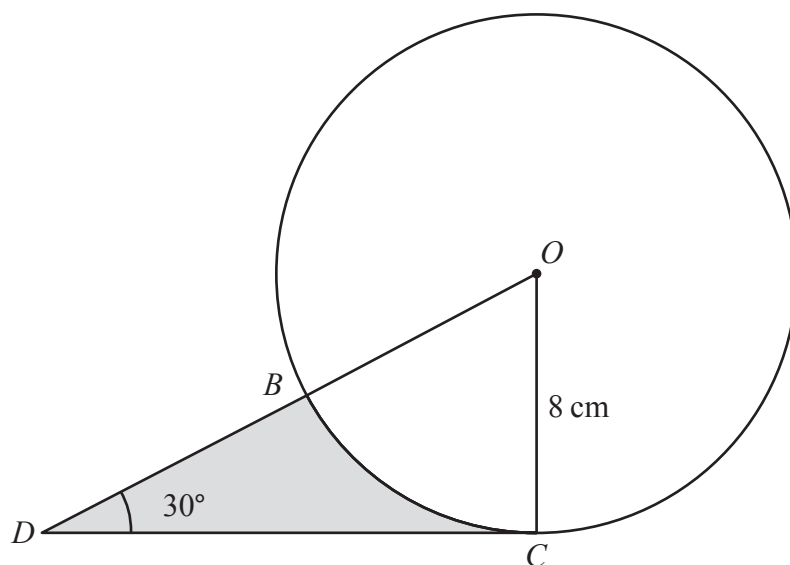
Show all your working.

..... cm

(Total for Question 21 is 5 marks)



Diagram **NOT**
accurately drawn



In the diagram B and C are points on a circle centre O

The point D is such that DBO is a straight line.

DC is the tangent to the circle at C

$$CO = 8 \text{ cm and } \angle ODC = 30^\circ$$

The region inside OCD but outside of the sector OBC is shaded.

Calculate the area, in cm^2 to 3 significant figures, of this shaded region.
Show all your working.

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..... cm^2

(Total for Question 22 is 5 marks)

23 (a) Simplify $w^3 \times w^{-5}$

.....
(1)

(b) Simplify $5x^0$

.....
(1)

(c) Simplify fully $(8a^3b^9)^{\frac{1}{3}}$

.....
(2)

(d) Simplify fully $\frac{(36c^4d^2)^{\frac{1}{2}}}{3c^{-5}d^3}$

.....
(3)

(Total for Question 23 is 7 marks)



P 8 1 3 3 2 A 0 1 9 2 4

24 Zaid has two bags of marbles, A and B

There are 10 marbles in bag A and 15 marbles in bag B

In bag A , there are 5 red marbles, 3 blue marbles and 2 yellow marbles.

In bag B , there are 6 red marbles, 2 blue marbles and 7 purple marbles.

Zaid takes at random one marble from bag A and one marble from bag B

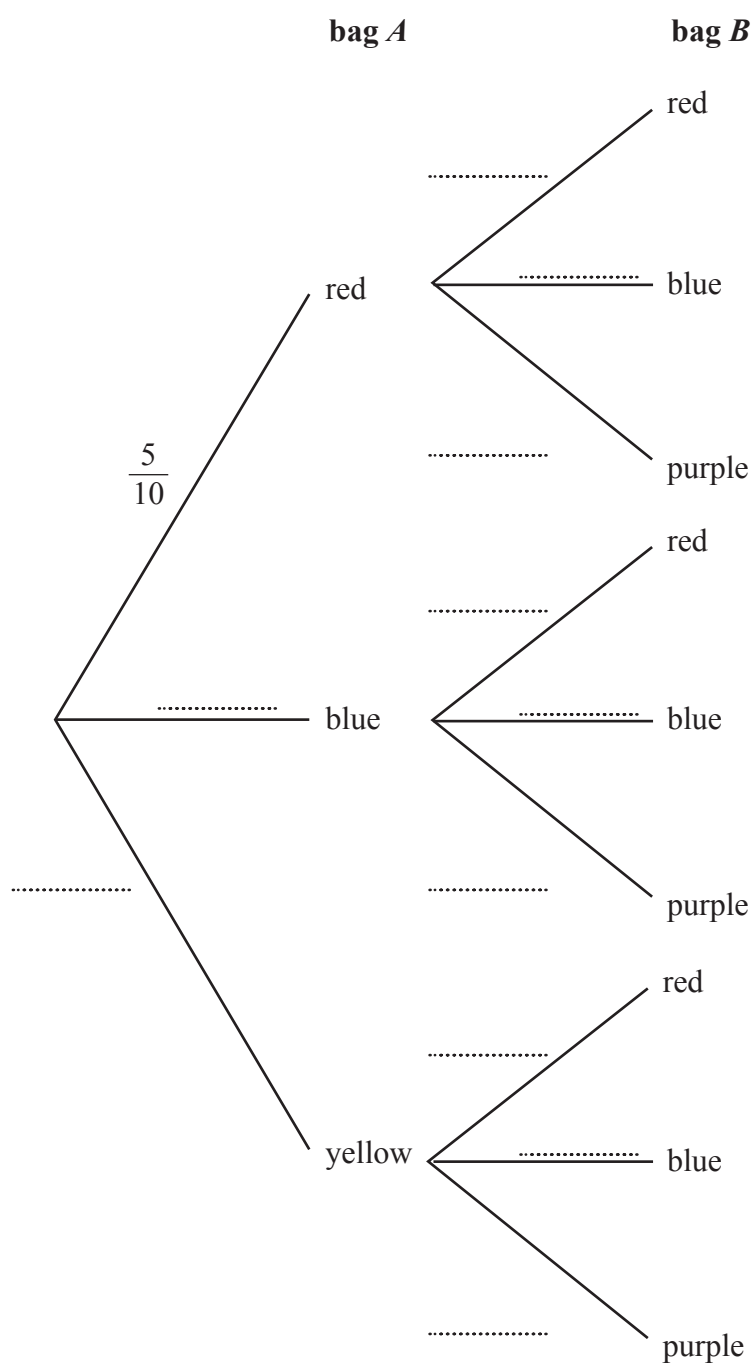
(a) Complete the probability tree diagram on the opposite page.

(3)

(b) Calculate the probability that the two marbles are **not** the same colour.

(3)





(Total for Question 24 is 6 marks)



25 A , X and Y are three points on horizontal ground.

AB is a vertical mast with height 45 m

X is 2000 m due west of A

The bearing of Y from A is 320°

The angle of elevation of B from Y is 10°

Calculate, to the nearest degree, the bearing of X from Y

Show all your working.



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(Total for Question 25 is 6 marks)

Turn over for Question 26



26 Solve the simultaneous equations

$$\begin{aligned}x^2 &= 2 - y^2 \\ 3y - x &= 4\end{aligned}$$

Show clear algebraic working.

(Total for Question 26 is 6 marks)

TOTAL FOR PAPER IS 100 MARKS

